CLI (Command Line Interface)

For the CLI, you can provide a Python script that renames a file via command-line arguments.

Python code:

```
def length_converter():
print("\nLength Conversion")
print("1. Kilometer to Meter")
print("2. Meter to Kilometer")
choice = input("Choose option (1/2): ")
value = float(input("Enter value: "))
if choice == "1":
result = value * 1000
print(f"Result: {result} meters")
elif choice == "2":
result = value / 1000
print(f"Result: {result} kilometers")
else:
print("Invalid choice!")

def mass_converter():
print("\nMass Conversion")
print("1. Kilogram to Gram")
print("2. Gram to Kilogram")
choice = input("Choose option (1/2): ")
value = float(input("Enter value: "))
if choice == "1":
result = value * 1000
print(f"Result: {result} grams")
elif choice == "2":
result = value / 1000
print(f"Result: {result} kilograms")
```

```
else:
print("Invalid choice!")

def temperature_converter():

print("\nTemperature Conversion")
print("1. Celsius to Fahrenheit")
print("2. Fahrenheit to Celsius")
choice = input("Choose option (1/2): ")
value = float(input("Enter value: "))
if choice == "1":
result = (value * 9/5) + 32
print(f"Result: {result} °F")
elif choice == "2":
result = (value - 32) * 5/9
print(f"Result: {result} °C")
else:
print("Invalid choice!")

def main():
while True:
print("\n===== UNIT CONVERTER SYSTEM =====")
print("1. Length Converter")
print("2. Mass Converter")
print("3. Temperature Converter")
print("4. Exit")
option = input("Select option (1-4): ")
if option == "1":
length_converter()
elif option == "2":
mass_converter()
```

```
elif option == "3":
temperature_converter()
elif option == "4":
print("Exiting program. Thank you!")
break
else:
print("Invalid selection. Try again.")


if __name__ == "__main__":
main()
```

OUTPUT:

```
apple@apples-MacBook-Pro CLI % python3 cli_unit_converter.py

===== UNIT CONVERTER SYSTEM =====
1. Length Converter
2. Mass Converter
3. Temperature Converter
4. Exit
Select option (1-4): 1

Length Conversion
1. Kilometer to Meter
2. Meter to Kilometer
Choose option (1/2): 1
Enter value: 7.5
Result: 7500.0 meters

===== UNIT CONVERTER SYSTEM =====
1. Length Converter
2. Mass Converter
3. Temperature Converter
4. Exit
Select option (1-4): 4
Exiting program. Thank you!
```

GUI (Graphical User Interface):

```
import tkinter as tk
from tkinter import ttk, messagebox


def convert():
try:
value = float(entry_value.get())
category = combo_category.get()
```

```
conversion = combo_conversion.get()
if category == "Length":
if conversion == "Kilometer to Meter":
result = value * 1000
output_label.config(text=f"Result: {result} meters")
elif conversion == "Meter to Kilometer":
result = value / 1000
output_label.config(text=f"Result: {result} kilometers")

elif category == "Mass":
if conversion == "Kilogram to Gram":
result = value * 1000
output_label.config(text=f"Result: {result} grams")
elif conversion == "Gram to Kilogram":
result = value / 1000
output_label.config(text=f"Result: {result} kilograms")
elif category == "Temperature":
if conversion == "Celsius to Fahrenheit":
result = (value * 9/5) + 32
output_label.config(text=f"Result: {result} °F")
elif conversion == "Fahrenheit to Celsius":
result = (value - 32) * 5/9
output_label.config(text=f"Result: {result} °C")
else:
messagebox.showerror("Error", "Invalid selection!")
except ValueError:
messagebox.showerror("Error", "Please enter a valid number!")

def update_options(event):
category = combo_category.get()
if category == "Length":
```

```
combo_conversion['values'] = (
"Kilometer to Meter",
"Meter to Kilometer"
)
elif category == "Mass":
combo_conversion['values'] = (
"Kilogram to Gram",
"Gram to Kilogram"
)
elif category == "Temperature":
combo_conversion['values'] = (
"Celsius to Fahrenheit",
"Fahrenheit to Celsius"
)
combo_conversion.current(0)

root = tk.Tk()
root.title("Unit Converter System")
root.geometry("400x300")
root.resizable(False, False)

tk.Label(root, text="UNIT CONVERTER SYSTEM",
font=("Arial", 14, "bold")).pack(pady=10)

tk.Label(root, text="Enter Value:").pack()
entry_value = tk.Entry(root, width=20)
entry_value.pack(pady=5)

tk.Label(root, text="Select Category:").pack()
combo_category = ttk.Combobox(root, state="readonly",
values=("Length", "Mass", "Temperature"))
```

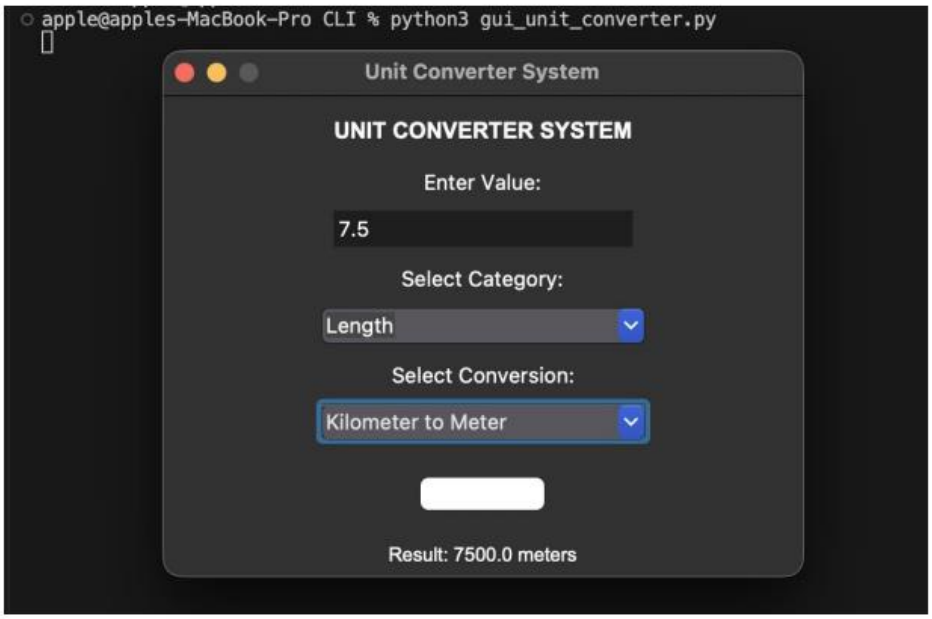
```
combo_category.pack(pady=5)
combo_category.bind("<<ComboboxSelected>>", update_options)

tk.Label(root, text="Select Conversion:").pack()
combo_conversion = ttk.Combobox(root, state="readonly")
combo_conversion.pack(pady=5)

tk.Button(root, text="Convert",
command=convert,
bg="green",
fg="white").pack(pady=10)

output_label = tk.Label(root, text="Result: ",
font=("Arial", 12))
output_label.pack(pady=10)
root.mainloop()
```

OUTPUT:



3.VOICE USER INTERFACE:

```
import sounddevice as sd
from scipy.io.wavfile import write
import speech_recognition as sr
fs = 44100 # Sample rate
seconds = 5 # Recording duration

def record_audio():
print("Speak now...")
recording = sd.rec(int(seconds * fs), samplerate=fs, channels=1)
sd.wait()

write("voice.wav", fs, recording)
print("Recording complete.")

def convert_units(command):
command = command.lower()
try:
words = command.split()
value = float(words[0])
from_unit = words[1]
to_unit = words[3]
# Length
if from_unit in ["kilometer", "kilometers"] and to_unit in ["meter", "meters"]:
result = value * 1000
elif from_unit in ["meter", "meters"] and to_unit in ["kilometer", "kilometers"]:
result = value / 1000
# Weight
elif from_unit in ["kilogram", "kilograms"] and to_unit in ["gram", "grams"]:
result = value * 1000
elif from_unit in ["gram", "grams"] and to_unit in ["kilogram", "kilograms"]:
result = value / 1000
```

```
else:
return "Conversion not supported."
return f"The result is {result} {to_unit}"
except:
return "Please say like: 10 kilometer to meter"

def recognize_speech():
r = sr.Recognizer()
with sr.AudioFile("voice.wav") as source:
audio = r.record(source)
try:
text = r.recognize_google(audio)
print("You said:", text)
return text
except:

return "Could not understand."

def main():
print("=== Real Voice Unit Converter (No PyAudio) ===")
print("You have 5 seconds to speak.\n")
while True:
record_audio()
command = recognize_speech()
if command.lower() == "exit":
print("Goodbye!")
break
result = convert_units(command)
print("Assistant:", result)

if __name__ == "__main__":
```

main()

OUTPUT:

```
apple@apples-MacBook-Pro CLI % source /Users/apple/CLI/venv/bin/activate
(venv) apple@apples-MacBook-Pro CLI % python3 vui_unit_converter.py
=== Real Voice Unit Converter (No PyAudio) ===
You have 5 seconds to speak.

Speak now...
```